

Section 2F. Bias-Variance Tradeoff

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Bias-Variance Tradeoff

We can explain the shape of the test MSE as follows:

- ▶ Consider a training dataset $\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where the input/output data pairs $(\mathbf{x}_i, y_i)$ are drawn independently from an additive model:

$$\mathbf{x}_i \sim f_X \text{ and } y_i = f(\mathbf{x}_i) + \varepsilon \text{ with } \varepsilon \sim f_\varepsilon$$

where ε is a measurement noise and f is an unknown regression function that “nature” is using to generate the dataset. Notice that the additive model induces a joint PDF, f_{XY}

- ▶ Assuming a parametric form of our regression function $\hat{f}(\mathbf{x}; \theta)$, our task is to find the parameters θ^* that minimize the *training MSE*

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Tr}}} \left(y_i - \hat{f}(\mathbf{x}_i; \theta) \right)^2$$

Bias-Variance Tradeoff (cont.)

- Once we have a trained model $\hat{f}(\mathbf{x}; \theta^*)$, we are interested in analyzing the *test MSE*. Considering a new datapoint $(\mathbf{x}_0, y_0) \sim f_{XY}$ (not used in the training process), we can *theoretically* write the test MSE as

$$\text{MSE}_{\text{Te}} = \mathbb{E}_{(\mathbf{x}_0, y_0) \sim f_{XY}} \left[\left(y_0 - \hat{f}(\mathbf{x}_0; \theta^*) \right)^2 \right]$$

- Notice that, since $\mathbf{x}_0$ is a r.v., the function $\hat{f}(\mathbf{x}_0; \theta^*)$ is also a r.v. The test MSE can be written as (proof omitted)

$$\text{MSE}_{\text{Te}} = \text{Var} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] + \left(\text{Bias} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] \right)^2 + \text{Var} [\varepsilon]$$

where

$$\text{Bias} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] = \mathbb{E} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] - f(\mathbf{x}_0)$$

$$\text{Var} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] = \mathbb{E} \left[\left(\hat{f}(\mathbf{x}_0; \theta^*) - \mathbb{E} \left[\hat{f}(\mathbf{x}_0; \theta^*) \right] \right)^2 \right]$$

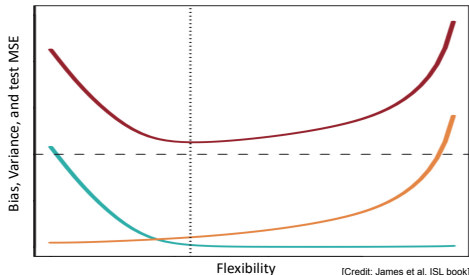
Bias-Variance Tradeoff (cont.)

- ▶ The sum of the terms $\text{Var} [\hat{f}(\mathbf{x}_0; \theta^*)] + \left(\text{Bias} [\hat{f}(\mathbf{x}_0; \theta^*)] \right)^2$ is called the *reducible error*, since it can be reduced by choosing a good parametric function $\hat{f}(\mathbf{x}; \theta)$
- ▶ The term $\text{Var} [\varepsilon]$ is called *irreducible error*, since it is always there, even when your parametric function is exactly the same as the regression function f that “Nature” is using to generate the dataset you observe

Bias-Variance Tradeoff (cont.)

The two terms in the reducible error depend on the flexibility of the model $\hat{f}(\mathbf{x}; \theta)$

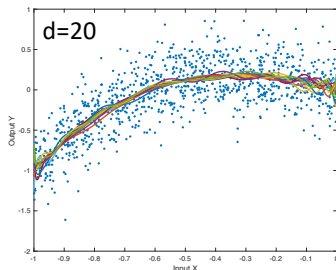
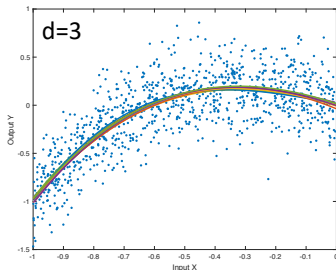
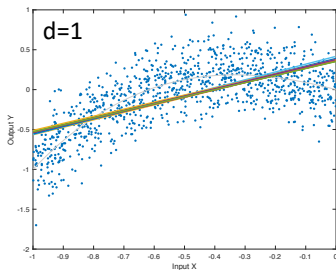
- ▶ The *bias term* $\text{Bias}[\hat{f}(\mathbf{x}_0; \theta^*)]$ decreases monotonically as the flexibility of the model increases (cyan plot)
- ▶ The variance term $\text{Var}[\hat{f}(\mathbf{x}; \theta)]$, increases monotonically as the flexibility of the model increases (orange plot)

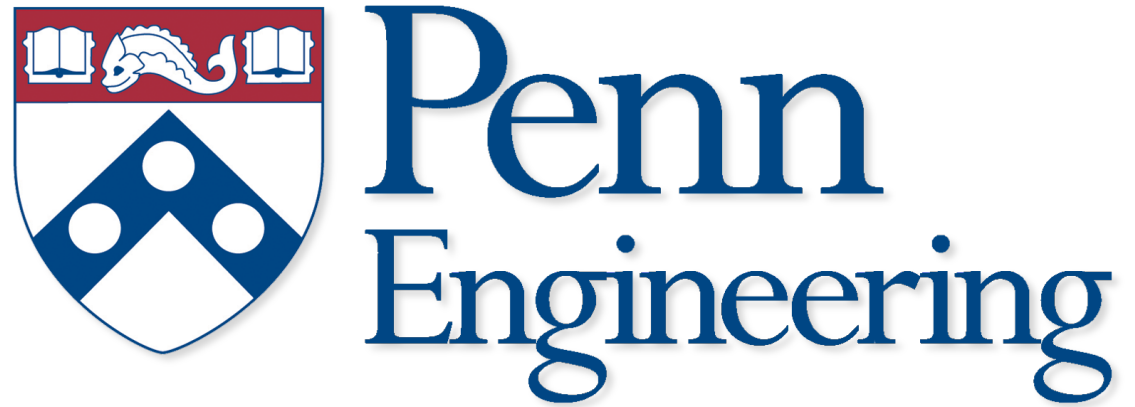


Bias-Variance Tradeoff (cont.)

Numerical interpretation: Run 10 different polynomial fits (with 10 different training datasets) when $d = 1, 3$ and 20 .

- ▶ For $d = 1$ (rigid case), the variance is low, but the bias is high
- ▶ For $d = 3$ (cubic case), the variance and bias are both low
- ▶ For $d = 20$ (flexible case), the bias is still low, but the variance increases





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